

Homework 3

Due Date: End of Week 4, Sunday at midnight

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Q1: Suppose you are on a game show, and you're given the choice of three doors numbered 1, 2, and 3. Behind one of these doors is a new car and there behind the other two there is nothing. You initially pick door 1. Before the game, the show's host reveals what is behind door 1, she opens door 2 and shows you this door has nothing behind it. She then asks "Would you like to stay with door 1 or switch to door 3?". Is it advantageous to switch from door 1 to door 3 given door 2 has been shown to have nothing? (Hint: properly defining each event, conditional probability, Bayes' Rule will be useful here)

Q2: How many people are needed in a room such that there is at least a 50% chance that at least 2 people have the same birthday? Assume each person has an equal and mutually independent likelihood of a birthday on a given day ($p = 1/365$ for any of the n people in the room). (Hint: If A and B are mutually exclusive events, then $P(A) = 1 - P(B)$).

Q3: Suppose we are interested in using linear regression to model bilateral trade flows between nations. Index each country pair as i, j . Define Y_{ij} as the bilateral flow between country i and j , X_i and X_j as GDP for countries i and j , D_{ij} as the distance between the two countries, G a constant, and U_{ij} as any other unobserved (in terms of the model) influences affecting Y_{ij} . The model is specified as follows:

$$Y_{ij} = G \frac{X_i X_j}{D_{ij}} \cdot U_{ij}$$

(a) Describe an approach to estimate this non-linear model with linear regression. Note that you might need to slightly modify the model given above. Justify any transformations or added terms.

(b) Using the linear specification derived above, give an interpretation of the constant term. Give an interpretation for the coefficients associated with the terms X_i and X_j . Give an interpretation of the coefficients associated with D_{ij} .

(c) Describe what data would be needed to estimate the model. Are there any desirable properties of such a sample? (Hint: any conditions on n ? They can be general)

(d) Suppose the model above is missing an important factor in determining bilateral trade flows (e.g. common languages, trade agreements between i and j). Where will the effects of this factor show up in the model specified above? Discuss how that should change the interpretation of the estimated coefficients for this specification.